

FIN-220 Hands-On Machine Learning
HW 1
Due on September 22, 2025

1. What is a labeled training set?
2. What type of algorithm would you use to allow a robot to walk in various unknown terrains?
3. What type of algorithm would you use to segment your customers into multiple groups?
4. Would you frame the problem of spam detection as a supervised learning problem or an unsupervised learning problem?
5. What is out-of-core learning?
6. What is the difference between a model parameter and a model hyperparameter?
7. What do model-based algorithms search for? What is the most common strategy they use to succeed? How do they make predictions?
8. If your model performs great on the training data but generalizes poorly to new instances, what is happening? Can you name three possible solutions?
9. What is the purpose of a validation set?
10. What is the train-dev set, when do you need it, and how do you use it?
11. What can go wrong if you tune hyperparameters using the test set?
12. Explain whether each scenario is a classification or regression problem, and indicate whether we are most interested in inference or prediction. Finally, provide n (number of sample points, observations) and p (number of features).
 - We collect a set of data on the top 500 firms in the US. For each firm we record profit, number of employees, industry and the CEO salary. We are interested in understanding which factors affect CEO salary.
 - We are considering launching a new product and wish to know whether it will be a success or a failure. We collect data on 20 similar products that were previously launched. For each product we have recorded whether it was a success or failure, price charged for the product, marketing budget, competition price, and ten other variables.

- We are interested in predicting the % change in the USD/Euro exchange rate in relation to the weekly changes in the world stock markets. Hence we collect weekly data for all of 2012. For each week we record the % change in the USD/Euro, the change in the US market, the % change in the British market, and the % change in the German market.
13. Read the paper *Scaling to Very Very Large Corpora for Natural Language Disambiguation* by Banko and Brill and write a brief summary.
 14. The table below provides a training data set containing six observations, three predictors, and one qualitative response variable.

Obs.	X_1	X_2	X_3	Y
1	0	3	0	Red
2	2	0	0	Red
3	0	1	3	Red
4	0	1	2	Green
5	-1	0	1	Green
6	1	1	1	Red

Suppose we wish to use this data set to make a prediction for Y when $X_1 = X_2 = X_3 = 0$ using K-nearest neighbors.

- Compute the Euclidean distance between each observation and the test point, $X_1 = X_2 = X_3 = 0$.
- What is our prediction with $K = 1$? Why?
- What is our prediction with $K = 3$? Why?